Exam 1 Recap (review slides)

Module 1 - Introduction & Math Review · Review set · 9 Sep 2026

Name _____ Date _____

A solution of a drug was freshly prepared at a concentration of 300 mg/mL. After 30 days, the drug concentration in the solution was 80 mg/mL.

Module 1 practice, item 1

Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

Module 1 practice, item 2

Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?

Module 3 practice, item 1

Recommend an infusion rate to achieve a steady-state concentration of 24 mg/L of an agent that has an elimination half-life of 5 hours and a volume of distribution of approximately 22 L.

Module 3 practice, item 2

Recommend a loading dose that will achieve a steady-state concentration of 24 mg/L for an agent whose volume of distribution is approximately 22 L.

Module 3 practice, item 3

The same agent, with an elimination half-life of 5 hours and a volume of distribution of 22 L, is infused at the rate that gives a steady-state concentration of 24 mg/L. What is the expected plasma concentration 10 hours after the start of the infusion?

Module 3 practice, item 4

An agent with an elimination half-life of 5 hours and a volume of distribution of 22 L is infused at the rate that would give a steady-state concentration of 24 mg/L. The infusion is stopped 5 hours after it was started. What is the plasma concentration 3 hours after cessation?